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March 3, 2026

Abstract

Knowledge tracing is a core component of adaptive learning systems, yet much of the literature assumes binary correctness as the observation signal. In many educational settings, however, learner evidence is polytomous, including ordered rubric levels and partial credit, where both ordinal structure and interpretability matter for measurement and intervention.

We propose DEEP-MIRT, a modular framework that extends knowledge tracing to polytomous responses by bridging sequential neural modeling with psychometric Item Response Theory. The framework couples a swappable sequential backbone with an interchangeable response head, separating the question of how learner ability evolves over time from the question of how that ability maps to ordered response categories. Instantiated with a Generalized Partial Credit Model (GPCM) head, DEEP-MIRT produces psychometrically interpretable outputs at every interaction, including an evolving student ability estimate, item discrimination values, and ordered difficulty thresholds. The model delivers strong predictive performance while accurately recovers ground-truth item parameters across varying category resolutions. The framework advances knowledge tracing from a purely predictive instrument into a measurement-aware system that grounds instructional decision-making in interpretable psychometric theory.

Introduction

Educational assessments often target graded performance rather than pass/fail outcomes. Rubric-based scoring, partial-credit items, and constructed-response tasks encode degrees of correctness that binary labels collapse into a single signal (Black and Wiliam, 1998; Heritage, 2007). While automated scoring systems can deliver this ordinal structure at scale in intelligent tutoring contexts (Shute, 2011; Baral et al., 2021), most KT benchmarks, evaluation protocols, and modeling paradigms still assume binary correctness. This leaves ordinal outcomes comparatively underexplored in KT, even though they are central to authentic assessment practice.

Beyond the outcome type, interpretability remains a key concern. Most KT models learn high-dimensional states optimized for prediction, but these states are not, by construction, tied to interpretable quantities. Therefore, constructs that educators and assessment designers routinely work with, such as student ability level, item discriminatory power, or the difficulty steps between adjacent rubric categories, are not directly

identifiable. Item Response Theory (IRT) provides exactly this structure through identifiable parameters including student ability, item discrimination, and ordered step difficulties, grounding score interpretation in established psychometric theory (Muraki, 1997; Embretson and Reise, 2000; Messick, 1995). Standard IRT models, however, treat ability as fixed and are not designed for sequential interaction data. This work therefore extends KT to polytomous responses by incorporating the interpretable parameter structure of IRT within a dynamic sequential framework.

Three lines of existing work address these issues in isolation, leaving a gap at their intersection. First, deep KT models, from Deep Knowledge Tracing (DKT; Piech et al., 2015) to memory-augmented and attention-based variants (Zhang et al., 2017; Pandey and Karypis, 2019), capture temporal learning dynamics but typically assume dichotomous correctness and learn latent states with limited psychometric interpretability. Second, polytomous Item Response Theory (IRT) models such as the Generalized Partial Credit Model (GPCM; Muraki, 1997) provide principled, interpretable measurement of ordered responses with identifiable parameters (student ability, item discrimination, ordered step thresholds), but they generally treat ability as time-invariant and are not designed for sequential interaction logs. Third, recent efforts that inject IRT structure into deep KT (Yeung, 2019; Converse et al., 2021) improve interpretability but largely remain restricted to dichotomous outcomes. An early partial exception is Wang and Heffernan (2013), who extended Bayesian Knowledge Tracing to allow partial credit via continuous Bayesian nodes, but without deep sequential modeling or recovering standard polytomous IRT step structure. Yet these approaches has not yet reconciled sequential dynamics with ordinal response modeling while retraining psychometric identifiability.

We propose *DEEP-MIRT*, a modular framework that extends KT to Polytomous Knowledge Tracing (PKT) by coupling a swappable sequential backbone with an IRT-grounded response head. In this setting, PKT is the task of predicting the full distribution over ordered response categories from a learner's interaction history. We instantiate the framework with a GPCM response head, yielding DEEP-GPCM, which combines a Dynamic Key-Value Memory Network (DKVMN) backbone with a GPCM observation model. At each timestep, the GPCM head maps the learner state to a distribution over ordinal categories and exposes psychometrically interpretable components, including an evolving student ability estimate, as well as learned item parameters such as per-item discrimination values and ordered step difficulties corresponding to rubric level transitions. While the experiments focus on ordinal PKT, the framework accommodates other polytomous formats through alternative response heads.

Our main contributions are:

1. **A modular framework for polytomous knowledge tracing.** We formulate PKT as predicting the full ordinal response distribution from interaction history and propose DEEP-MIRT, which decouples a sequential backbone (encoder, memory, summary) from an interchangeable psychometric response head. The head interface accepts student ability θ_t and item identity $\mathbf{k}_t$ and returns a category probability vector; any differentiable observation model satisfying this contract is a valid instantiation. This design separates the educational measurement question (which model governs how ability maps to rubric scores?) from the knowl-

edge tracing question (how does ability evolve over time?).

2. **GPCM instantiation with interpretable psychometric parameters.** We instantiate DEEP-MIRT with a GPCM-based response head as DEEP-GPCM, incorporating a monotonic threshold parameterization that enforces rubric ordering by construction. The resulting model produces evolving ability trajectories, item discrimination estimates, and ordered step difficulty thresholds, all of which correspond to constructs that educators and assessment designers already use in rubric design and standard-setting practice.

3. **Scalable item encoding and empirical evaluation.** We propose Static Item Embedding (SIE) for large item banks and evaluate ordinal prediction quality, psychometric parameter recovery, ablation of design components, and backward compatibility with binary KT benchmarks.

We investigate the following research questions:

- **RQ1 (Prediction).** Does coupling a DKVMN backbone with an ordinal, psychometrically constrained response head improve polytomous next-response prediction, and does it better preserve ordinal structure than format-unaware alternatives?

- **RQ2 (Learner State).** Do the inferred student ability trajectories θ_t exhibit psychologically plausible learning dynamics, and can DEEP-GPCM reveal meaningful learner patterns (mastery, growth, persistent difficulty) that are not visible from format-unaware latent state representations?

- **RQ3 (Parameter Recovery).** Do the IRT parameters inferred by DEEP-GPCM faithfully recover known ground-truth values from synthetic data, and are item parameter estimates stable across student subpopulations as assessed by split-half reliability?

- **RQ4 (Scalability).** Does Static Item Embedding maintain ordinal prediction quality and parameter recovery at large item-bank scales ($Q \sim 5000$) while substantially reducing parameter count compared to dense encoding strategies?

- **RQ5 (Ecological Validity).** When ordinal response categories are constructed from proxy signals in existing binary KT benchmarks (e.g., attempt counts), does the ordinal GPCM head yield measurable gains over format-unaware alternatives, and how sensitive are these gains to the quality of the proxy ordinal construction?

Related Work

Knowledge Tracing: From Binary to Polytomous

Knowledge tracing has evolved through several modeling paradigms, but most of them assume a binary mastery state. Bayesian Knowledge Tracing (BKT) models mastery of

a Knowledge Component (KC) as a two-state Hidden Markov Model (Corbett and Anderson, 1994). Deep Knowledge Tracing (DKT) reframed the task as sequence modeling with recurrent neural networks, learning high-dimensional latent states from interaction sequences (Piech et al., 2015). Dynamic Key-Value Memory Networks (DKVMN) introduced a structured memory that separates static concept representations (key memory) from evolving mastery states (value memory), offering a more transparent mapping between knowledge components and latent state than generic RNNs (Graves et al., 2016; Zhang et al., 2017). Attention-based models such as SAKT and AKT apply self-attention to weigh the relevance of prior interactions for next-step prediction (Pandey and Karypis, 2019; Ghosh et al., 2020), and recent work has further explored Transformer-based architectures (Choi et al., 2020; Liu et al., 2023). Throughout this progression, the response format has remained dichotomous: every model cited above predicts a binary correct/incorrect outcome.

Recent KT research has begun to acknowledge that correctness alone is insufficient. Lu et al. (2024) fuse process features (attempts, response time, historical accuracy) to enrich difficulty representations, and Li et al. (2025) integrate problem complexity and state-stability mechanisms to reduce noisy updates. These efforts treat richer signals as auxiliary features within a binary prediction framework. They do not change the response format itself, but they highlight that educational interactions carry information beyond pass/fail: precisely the information that ordinal and polytomous responses encode directly. The only prior work to move beyond binary responses in a KT context is Wang and Heffernan (2013), who extended BKT to partial credit using continuous Bayesian nodes, but without deep sequential modeling or IRT-based parameterization.

Interpretability in Deep Knowledge Tracing

In most DKT models, learned states are computed as high dimensional representations than psychometrically meaningful constructs (National Research Council, 2001). Several efforts address this gap by coupling neural architectures with Item Response Theory (IRT) structure. Deep-IRT attaches an IRT-style readout to DKVMN, exposing student ability and item difficulty parameters for each prediction (Yeung, 2019). DKVMN-MRI similarly integrates IRT within an LSTM-based memory framework (Xu et al., 2024). Converse et al. (2021) propose a general approach for incorporating IRT parameters into KT training. These IRT-augmented models improve interpretability: a teacher or assessment specialist can inspect the estimated ability and difficulty values, rather than relying on opaque hidden states. In educational terms, "interpretability" here means that model outputs correspond to constructs that practitioners already use (student proficiency levels, item difficulty, and discrimination quality), supporting the kinds of validity arguments that responsible assessment requires (Messick, 1995; Kane, 2013).

A critical limitation, however, is that all existing IRT-KT integrations operate on dichotomous responses. They estimate a single ability and difficulty for binary outcomes and do not model the ordered step structure of rubric scores or partial-credit items. Attention-based models provide a complementary form of interpretability through attention weights (which interactions were most relevant to a prediction), but they too lack psychometric parameterization and ordinal structure awareness.

Polytomous Item-Response Theory

Psychometrics provides a family of models for ordered responses grounded in a latent ability parameter θ and item-specific properties (Hambleton and Swaminathan, 1985; Embretson and Reise, 2000). Two traditions dominate for ordinal outcomes. Cumulative models such as the Graded Response Model (GRM; Samejima, 1969) define boundary parameters between adjacent categories, modeling the probability of scoring at or above category k as a logistic function of θ. Adjacent-category models, including the Partial Credit Model (PCM; Masters and Wright, 1997) and the Generalized Partial Credit Model (GPCM; Muraki, 1997), parameterize each step transition directly: the GPCM characterizes each item by a discrimination parameter α_j controlling how sharply the item differentiates ability levels, and ordered step difficulty thresholds $\boldsymbol{\beta}_j = (\beta_{j,0}, \ldots, \beta_{j,K-2})$ representing the ability level at which each consecutive rubric transition becomes equally likely. Adjacent-category models are the natural fit for rubric scoring, where performance progresses sequentially through ordered levels (Embretson and Reise, 2000; Wilson, 2005). For unordered categories, the Nominal Response Model (NRM; Bock, 1972) provides an analogous framework.

In the context of knowledge tracing, these static measurement paradigms must be reformulated to capture sequential learning dynamics. Formally, at time step t, a student interacts with an item indicated by a one-hot vector $q_t \in \{0,1\}^Q$ over Q unique items, yielding an ordinal response $r_t \in \{0, 1, \ldots, K-1\}$ across K rubric levels. Defining an interaction as the tuple $x_t = (q_t, r_t)$ and the cumulative history as $\mathcal{X}_t = \{x_1, \ldots, x_t\}$, the objective of polytomous knowledge tracing is to predict the conditional distribution of the next response:

$$P(r_{t+1} = k \mid q_{t+1}, \mathcal{X}_t), \quad k \in \{0, \ldots, K-1\}. \tag{1}$$

This work therefore explores how the interpretable parameter structures of polytomous IRT might be adapted for sequential modeling. We treat it as a time-varying latent state conditioned on the interaction history $\mathcal{X}_t$, aiming to interpret dynamic tracing results with psychometric models.

Methodology

Framework Overview

DEEP-MIRT is a modular architecture with two separable concerns. The *backbone* handles the knowledge tracing problem: it encodes item–response interactions, maintains a dynamic memory of the student's evolving knowledge state, and produces a summary vector $\mathbf{f}_t$ representing current student state. The *response head* handles the measurement problem: it maps the backbone's student state $\mathbf{f}_t$ and item identity $\mathbf{k}_t$ to a probability distribution over response categories through a psychometric observation model. Any differentiable function $h(\mathbf{f}_t, \mathbf{k}_t) \rightarrow \Delta^{K-1}$ (a distribution over K categories) satisfying this interface is a valid DEEP-MIRT instantiation. The backbone is agnostic to which measurement model governs the head.

This separation is conceptually and practically helpful for tracking learning progressions. The backbone addresses the learning dynamics question of how student

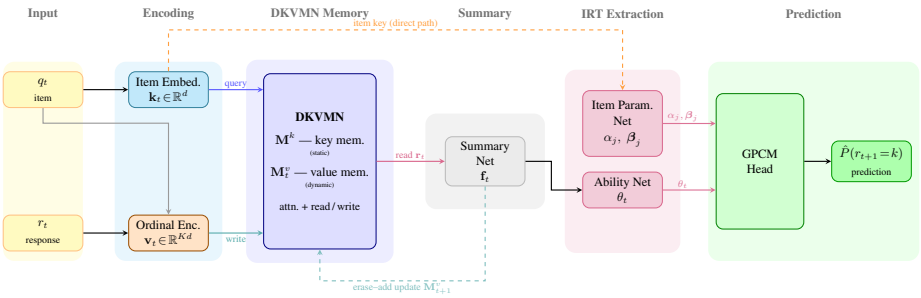

Figure 1: DEEP-MIRT framework at timestep t. Item parameters $(\alpha_j, \boldsymbol{\beta}_j)$ bypass the student state via the item embedding (dashed, top), consistent with the IRT assumption that discrimination and difficulty are student-independent. The GPCM head maps $(\theta_t, \alpha_j, \boldsymbol{\beta}_j)$ to the predicted ordinal distribution $\hat{P}(r_{t+1} = k)$.

ability evolves across a sequence of interactions, naturally handled by sequential neural architectures. The head addresses the measurement question of how current ability maps to the probability of each ordered rubric score, naturally handled by psychometric theory. Coupling them through a clean interface means the two concerns can be improved independently: a better backbone (e.g., Transformer instead of DKVMN) improves temporal modeling without changing the interpretable parameterization; a different head (e.g., GRM instead of GPCM) changes the measurement assumptions without retraining the backbone.

In this paper we instantiate the head with the GPCM over the other major polytomous families for three reasons: (1) it is an adjacent-category model, which is the natural match for rubric scoring where a student progresses sequentially through performance levels; (2) it produces a discrimination parameter α_j and ordered step difficulty thresholds $\boldsymbol{\beta}_j$ that correspond directly to rubric design decisions and can be inspected by assessment practitioners; (3) its monotone threshold constraint admits a clean differentiable parameterization (Section). A cumulative model such as the GRM (Samejima, 1969) would be a natural alternative for tasks where each scoring threshold represents an independent decision (e.g., analytic rubrics with independent sub-scores); a NRM (Bock, 1972) would apply when response categories are unordered (e.g., multiple-choice representing different misconceptions). We treat these as directions for future instantiation.

The backbone processes inputs through four sequential stages as shown in Figure 1: ordinal interaction encoding, DKVMN memory update, summary compression, and IRT-based parameter extraction. An ordinal-aware training objective supervises all stages jointly. The following subsections formalize each component.

Ordinal-Aware Interaction Representation

Linear Decay Encoding We encode an ordinal response r_t with a "triangular" kernel over ordered categories. Define a length-K ordinal code $\mathbf{s}_t \in \mathbb{R}^K$ by

$$\begin{aligned} s_t^{(k)} &= \max\left(0, 1 - \frac{|k - r_t|}{K-1}\right), \quad 0 \le k \le K-1, \\ \mathbf{s}_t &= [s_t^{(0)}; \ldots; s_t^{(K-1)}]. \end{aligned} \tag{2}$$

For example, with a $K = 4$ rubric and a response of $r_t = 2$ ("correct approach with errors"), the ordinal code is $\mathbf{s}_t = [0.33,\ 0.67,\ 1.0,\ 0.67]$, which spreads activation to neighboring categories while peaking at the observed level. This encoding preserves the ordinal structure: a response of 2 is represented as more similar to responses 1 and 3 than to response 0, reflecting the educational intuition that partial credit is closer to full credit than to no credit.

We then bind the ordinal code to the item indicator via a Kronecker product (equivalently, a block one-hot expansion):

$$\mathbf{x}_t = q_t \otimes \mathbf{s}_t \in \mathbb{R}^{KQ}. \tag{3}$$

This reduces to the standard binary interaction encoding when $K = 2$ (consistent with Zhang et al. (2017)). While this encoding is effective for moderately sized item banks, the $\mathcal{O}(KQ)$ dimensionality becomes a practical constraint for large Q; we address this in the following subsection.

Scalable Value Encoding for Large Item Banks Large-scale educational platforms maintain extensively large item banks with various grade levels. The triangular Kronecker encoding in (2) yields an interaction vector of dimension KQ, which grows linearly with item bank size. The subsequent value projection $\mathbf{v}_t = B^\top \mathbf{x}_t \in \mathbb{R}^{d_v}$ therefore requires a $KQ \times d_v$ weight matrix; at $Q = 5000$ and $K = 5$, this amounts to 1.6M parameters and a per-batch intermediate tensor of size $\mathrm{batch} \times S \times 25000$, placing substantial memory pressure on GPU-constrained deployment environments.

We propose **Static Item Embedding** (SIE), an alternative value encoding that decouples item identity from response pattern via two independent additive branches:

$$\mathbf{v}_t = W_{\text{item}}\, \mathbf{u}_{q_t} + W_{\text{resp}}\, \mathbf{s}_t + \mathbf{b}_{\text{resp}} \in \mathbb{R}^{d_v}, \tag{4}$$

where $\mathbf{u}_{q_t} \in \mathbb{R}^H$ is a *frozen* (non-trainable) unit-norm random vector assigned to item q_t at initialization, $\mathbf{s}_t \in \mathbb{R}^K$ is the triangular ordinal code from (2), and $W_{\text{item}} \in \mathbb{R}^{d_v \times H}$ and $W_{\text{resp}} \in \mathbb{R}^{d_v \times K}$ are learned projections shared across all items.

The factored structure in (4) separates two gradient channels: W_{item} is updated by item-identity signals, while W_{resp} is updated by response-pattern signals. Because item vectors are frozen, items cannot collapse to similar representations during training, a failure mode of learned separable embeddings at large Q (cf. Section). For two independent uniformly random unit vectors in $\mathbb{R}^H$, the expected squared inner product is $1/H$ (Dasgupta and Gupta, 2003), so pairwise interference concentrates around zero

as H grows; cross-item gradient crosstalk in W_{item} therefore decays with H independently of Q, making SIE well-suited for large item banks.

The frozen dimension H is set as the smallest power of two no less than $\max(128, \lfloor Q/2 \rfloor)$, clamped to $[128, 1024]$.[1] This schedule ensures near-orthogonality scales with item bank size while keeping the projection dimension bounded, in contrast to the $\mathcal{O}(KQ)$ complexity of the full Kronecker encoding.

Dynamic Key-Value Memory Network

We adopt the Dynamic Key-Value Memory Network (DKVMN) as the sequential backbone. Let $\mathbf{k}_t = A^\top q_t \in \mathbb{R}^{d_k}$ be a key embedding for the current item and $\mathbf{v}_t \in \mathbb{R}^{d_v}$ be a value embedding for the interaction, obtained via the Kronecker projection $\mathbf{v}_t = B^\top \mathbf{x}_t$ (ordinal-aware encoding) or the Static Item Embedding in (4) (scalable encoding). The model maintains a static key memory $\mathbf{M}^k \in \mathbb{R}^{N \times d_k}$ and a dynamic value memory $\mathbf{M}_t^v \in \mathbb{R}^{N \times d_v}$. Educationally, the key memory represents latent skill areas or knowledge components, the value memory tracks the student's evolving mastery of each area, and the attention weights $\mathbf{w}_t$ (below) indicate which skill areas a given item exercises. The summary vector $\mathbf{f}_t$ aggregates the student's current mastery state relevant to the item at hand.

Specifically, the key query is first projected via a learned transform, and attention weights over memory slots are computed as

$$\tilde{\mathbf{k}}_t = \tanh(W_{\text{attn}} \mathbf{k}_t), \qquad \mathbf{w}_t = \text{softmax}(\mathbf{M}^k \tilde{\mathbf{k}}_t) \in \mathbb{R}^N, \qquad w_t(i) = [\mathbf{w}_t]_i, \quad (5)$$

and the read vector is

$$\mathbf{r}_t = \sum_{i=1}^{N} w_t(i)\, \mathbf{M}_t^v(i) \in \mathbb{R}^{d_v}. \quad (6)$$

We fuse item context and retrieved state into a summary vector

$$\mathbf{f}_t = \tanh(W_{\mathbf{f}} [\mathbf{r}_t; \mathbf{k}_t] + b_{\mathbf{f}}), \quad (7)$$

which is then used by the response head to predict r_{t+1}.

Following memory-augmented neural networks, the value memory is updated via erase-add operations:

$$\mathbf{e}_t = \sigma(W_{\mathbf{e}} \mathbf{v}_t + b_{\mathbf{e}}), \quad (8)$$

$$\mathbf{a}_t = \tanh(W_{\mathbf{a}} \mathbf{v}_t + b_{\mathbf{a}}), \quad (9)$$

$$\mathbf{M}_{t+1}^v(i) = \mathbf{M}_t^v(i) \big[1 - w_t(i) \mathbf{e}_t\big] + w_t(i) \mathbf{a}_t. \quad (10)$$

DEEP-GPCM: an Ordinal IRT Head

Instead of the binary readout (e.g., $\sigma(W_p^\top \mathbf{f}_t + b_p)$), we use a Generalized Partial Credit Model (GPCM) head to obtain category probabilities for ordered responses. Let j denote the item exercised at time t (i.e., $[q_t]_j = 1$). We compute a time-varying ability

[1] Concretely, $H = 128$ for $Q \leq 256$, $H = 256$ for $Q \approx 500$, $H = 512$ for $Q \approx 1000$, and $H = 1024$ for $Q \geq 2000$.

proxy θ_t and item-specific parameters (thresholds and discrimination) as

$$\theta_t = W_\theta \mathbf{f}_t + b_\theta, \tag{11}$$

$$\beta_{j,0} = W_\beta \mathbf{k}_t + b_\beta, \tag{12}$$

$$\beta_{j,k} = \beta_{j,k-1} + \mathrm{softplus}(W_{g[k]}\mathbf{k}_t + b_{g[k]}), \quad 1 \le k \le K-2, \tag{13}$$

$$\alpha_j = \exp(W_\alpha \mathbf{k}_t + b_\alpha)\,. \tag{14}$$

The construction in (13) enforces $\beta_{j,0} < \beta_{j,1} < \cdots < \beta_{j,K-2}$ by adding strictly positive increments (via softplus) to each preceding threshold. This monotonicity guarantee has direct educational meaning: achieving rubric level $k{+}1$ always requires higher ability than achieving level k, respecting the ordering that a rubric designer intended. Without this constraint, the model could learn disordered thresholds where some rubric levels are never the most probable response at any ability level, a well-documented problem in educational measurement that signals a poorly functioning item (Wilson, 2005).

Because discrimination is an item-intrinsic property in the GPCM, α_j is derived solely from the item key $\mathbf{k}_t$, deliberately excluding the student-state summary $\mathbf{f}_t$. This separation ensures that the estimated discrimination reflects the item's inherent capacity to differentiate ability levels, independently of which student is currently being modelled — consistent with the IRT assumption that item parameters are population-invariant (Hambleton and Swaminathan, 1985). Item-level discrimination is recovered by averaging per-timestep estimates across all encounters with a given item (Meulders and Xie, 2004). The exponential activation ensures $\alpha_j > 0$.

Given $(\theta_t, \alpha_j, \{\beta_{j,k}\}_{k=0}^{K-2})$, the GPCM category probability is

$$p_{t,k} = \frac{\exp\Big(\sum_{i=0}^{k-1} \alpha_j(\theta_t - \beta_{j,i})\Big)}{\sum_{c=0}^{K-1} \exp\Big(\sum_{i=0}^{c-1} \alpha_j(\theta_t - \beta_{j,i})\Big)}. \tag{15}$$

At each timestep, the model evaluates this standard GPCM formulation (cf. Section) using the *dynamically estimated* student ability θ_t and item parameters $\alpha_j, \boldsymbol{\beta}_j$. Unlike classical GPCM estimation where ability is static, here θ_t evolves as new interactions are observed, enabling the model to track learning over time while retaining the interpretable measurement structure of polytomous IRT.

We predict $\hat{r}_{t+1} = \arg\max_k p_{t,k}$, and optimize the combined ordinal loss in (16) over the observed responses.

Training Objective for Ordinal Data

Because the GPCM defines category probabilities via (15) rather than cumulative links, we optimize a weighted categorical objective with an ordinal distance penalty to mitigate class imbalance (Gutiérrez et al., 2016). The loss combines two complementary components:

$$\mathcal{L} = \lambda_{\mathrm{focal}}\mathcal{L}_{\mathrm{focal}} + \lambda_{\mathrm{WOL}}\mathcal{L}_{\mathrm{WOL}}, \tag{16}$$

where $\mathcal{L}_{\mathrm{focal}}$ is the focal loss (Lin et al., 2020), which down-weights well-classified examples and focuses training on ambiguous responses near category boundaries. The

weighted ordinal loss $\mathcal{L}_{\text{WOL}}$ is

$$\mathcal{L}_{\text{WOL}} = \frac{1}{n} \sum_{i=1}^{n} \Big[(1 + \gamma |y_i - \hat{y}_i|) \cdot w_{y_i} \cdot (-\log \hat{p}_{i,y_i}) \Big], \tag{17}$$

where y_i is the true category, $\hat{p}_{i,y_i}$ is the predicted probability under (15), $\hat{y}_i = \arg\max_k \hat{p}_{i,k}$ is the predicted label, and γ controls the ordinal distance penalty. To balance classes, with c_k the sample count of class k, we set $w_k = \sqrt{\frac{n}{K \cdot c_k}}$, where n is the number of valid tokens in the batch.

Experiments

Datasets

Synthetic-Ordinal. To the best of our knowledge, there are no standardized public benchmarks for ordinal knowledge tracing. We therefore generate a synthetic dataset using an IRT-GPCM data generating process following established psychometric principles. This design provides controlled polytomous supervision $r_t \in \{0, \ldots, K-1\}$ and enables parameter recovery analysis with known ground truth.

We simulate 5000 students interacting with questions drawn from a pool of $Q = 200$ questions in random order. Each interaction sequence has variable length in $[50, 200]$. Students and questions are configured with *static* GPCM parameters following assumed priors:

- *Student ability:* $\theta \sim \mathcal{N}(0, 1)$,
- *Item discrimination:* $\alpha \sim \text{LogNormal}(0, 0.3)$,
- *Item thresholds:* $K-1$ values drawn independently from $\mathcal{N}(0, 1)$ and sorted to obtain $\beta_0 < \cdots < \beta_{K-2}$.

We generate four datasets with $K \in \{2, 3, 4, 5\}$ to test extensibility across category resolutions. Each dataset is split 80/20 for training/testing; a validation split is held out from training for early stopping and hyperparameter tuning.

Proxy-Ordinality To test our model on real educational logs, we derive ordinal labels from attempt counts in ASSISTments-style data (e.g., 1 attempt = mastery, 2–3 = partial, 4+ = struggle). Results and evaluation details are described in Section .

Binary Compatibility After validating the PKT instantiations on Synthetic-Ordinal, we confirm compatibility with binary KT. Due to implementation differences in batch handling and sequence-to-question lookup, we select two datasets:

- **ASSIST2015**[2]: 19,840 students and 100 KCs/questions.

[2]https://sites.google.com/site/assistmentsdata/datasets/2015-assistments-skill-builder-data

- **Synthetic-5**[3]: (Piech et al., 2015) simulated 4,000 students' responses with probability $p(r) = c + \frac{1}{1+\exp(-(\theta-\beta))}$ where $c = 0.25$ models question guessing. Each student is administered 50 questions in sequential order.

Models and Baselines

We compare DEEP-GPCM against four baselines that progressively relax psychometric and ordinal structure, each isolating a different design question:

- **DKVMN + Softmax**: DKVMN backbone with a K-way softmax head trained with categorical cross-entropy. No ordinal structure, no IRT constraints. Tests whether psychometric parameterization provides gains over a purely neural polytomous baseline.
- **DKVMN + Ordinal**: DKVMN backbone with a linear output head trained using the weighted ordinal loss (17). Introduces ordinal awareness without IRT parameterization, isolating the contribution of the GPCM structure over simpler ordinal objectives.
- **Static GPCM**: Standard GPCM fit from interaction data with student ability held fixed across items. No sequential dynamics. Represents the psychometric status quo and quantifies what sequential modeling adds beyond classical IRT estimation.
- **Dynamic GPCM**: GPCM with per-item α/β lookup tables and a gated recurrent ability update driven by prediction error at each step. No DKVMN memory. Isolates the contribution of sequential ability updating from DKVMN's distributed memory.
- **DEEP-GPCM (ours)**: DKVMN backbone with GPCM head and combined ordinal loss (16). Uses SIE encoding by default; the choice of encoding strategy is ablated in Section .

Prediction Mapping

For $K = 2$, predicted probabilities are converted to labels with a threshold of 0.5 following Zhang et al. (2017). For $K > 2$, we use $\hat{y}_t = \arg\max_k \hat{p}_{t,k}$ to map categorical probabilities to prediction labels.

Training Setup

We use the combined loss in (16) to handle class imbalance and ordinal agreement, with tunable weights λ_*.

We set $d_k = 64$ and $d_v = 128$ for key/value embedding dimensions and configure $N = 50$ memory slots for both memory matrices.

All models are trained with batch size 64 using Adam with learning rate 10^{-3}, reduced on performance plateau as per Mukherjee et al. (2019) for late-epoch stability.

[3]https://github.com/chrispiech/DeepKnowledgeTracing/tree/master/data/synthetic

We apply dropout 0.05 and early stopping if validation loss does not improve for 10 epochs. Models are implemented in PyTorch and trained on an RTX 4060 Laptop GPU.

Evaluation Metrics

We use the following metrics:

- **Categorical Accuracy (ACC)**: percentage of exact matches between predicted category and ground truth. While straightforward, ACC does not distinguish near-misses from far-misses and is therefore insufficient as a sole ordinal metric.
- **Quadratic Weighted Kappa (QWK)** (Cohen, 1968): our primary metric for ordinal tasks. QWK applies a quadratic penalty proportional to the squared distance between predicted and true categories, making it the standard agreement measure in automated essay scoring and rubric evaluation. A QWK of 0.0 indicates chance-level agreement; 1.0 indicates perfect agreement.
- **Kendall's Tau (τ)**: measures rank-order agreement between predicted and true response sequences, penalizing all inversions equally regardless of distance.
- **Area Under the ROC Curve (AUC)**: used for the binary case ($K = 2$) to maintain comparability with prior KT literature.

IRT Analysis Setup

For parameter recovery analysis on Synthetic-Ordinal, we use:

- $\boldsymbol{\theta}$ **Aggregation**: we take the mean ability estimate $\bar{\theta} = \frac{1}{T}\sum_{t=1}^{T}\theta_t$ across all timesteps for each student, averaging out sequence-position noise while exploiting the full interaction history.
- $\boldsymbol{\alpha}, \boldsymbol{\beta}$ **Aggregation**: because questions appear with variable frequency due to random sequencing, we aggregate item parameters based on occurrences and use the sample mean for recovery analysis.

We normalize learned parameters and true parameters using the priors used to generate each dataset. Specifically, both θ and β use z-score normalization. α is log-transformed and standardized with log-mean and log-standard deviation; we then rescale by 0.3 before the final exponential projection to match the generating scale.

Results

Ordinal Prediction on Synthetic-Ordinal (RQ1)

Table 1 reports prediction performance on Synthetic-Ordinal for $K \in \{2, 3, 4, 5\}$, addressing **RQ1**.

Two findings stand out. First, the τ advantage of DEEP-GPCM over the DKVMN baselines is largest at $K = 3$ and shrinks as K grows, reaching near-parity at $K = 5$. This is counterintuitive in terms of GPCM's ordinal structure where more thresholds

Table 1: Prediction performance on Synthetic-Ordinal ($Q = 200$) across $K \in \{2, 3, 4, 5\}$. Best per column in bold.

(a) $K = 2$ (Binary)

Model	**Acc(%)**	**AUC**
Static GPCM	61.6	0.664
Dynamic GPCM	68.5	0.753
DKVMN + Softmax	70.33	0.776
DKVMN + Ordinal	70.59	0.778
DEEP-GPCM (ours)	**71.4**	**0.779**

(b) $K = 3$

Model	**Acc(%)**	**QWK**	τ
Static GPCM	48.5	0.313	0.233
Dynamic GPCM	57.2	0.555	0.407
DKVMN + Softmax	60.18	0.570	0.407
DKVMN + Ordinal	59.59	0.592	0.431
DEEP-GPCM (ours)	**60.7**	**0.618**	**0.567**

(c) $K = 4$

Model	**Acc(%)**	**QWK**	τ
Static GPCM	39.0	0.322	0.244
Dynamic GPCM	49.6	0.644	0.490
DKVMN + Softmax	53.58	0.650	0.482
DKVMN + Ordinal	52.58	0.684	0.516
DEEP-GPCM (ours)	**53.7**	**0.710**	**0.627**

(d) $K = 5$

Model	**Acc(%)**	**QWK**	τ
Static GPCM	37.4	0.390	0.298
Dynamic GPCM	45.3	0.717	0.553
DKVMN + Softmax	**51.73**	0.724	0.541
DKVMN + Ordinal	50.13	**0.762**	**0.582**
DEEP-GPCM (ours)	50.7	0.759	0.576

should, if anything, help more at higher K. A feasible explanation is that at low K, the ordinal signal per observation is weak and the GPCM's ability-centred parameterization provides a stronger inductive bias for ranking students correctly; at high K, the richer per-step signal allows the DKVMN baselines to learn comparable rank structure from data alone. Second, the IRT ablation shows that sequential dynamics, not item parameterization, drive the bulk of performance: Dynamic GPCM, which shares item tables with Static GPCM but adds a single gated ability update, recovers most of the gap to the neural models. At $K = 4$ and $K = 5$, Dynamic GPCM nearly matches DKVMN + Softmax on QWK despite having no distributed memory, which suggests that a lightweight recurrent IRT model is a surprisingly competitive baseline when the response scale is fine-grained enough to provide a strong per-step learning signal.

Binary Compatibility on Standard KT Benchmarks

For practical deployment, it is important for PKT models to not sacrifice performance even when tested on binary data, since many existing item banks and datasets use dichotomous scoring, and a useful framework must handle both formats without architectural modification. We evaluate backward compatibility by instantiating DEEP-MIRT with $K = 2$ and testing on ASSIST2015 and Synthetic-5. Results are reported in Table 2. DEEP-GPCM achieves competitive performance compared to DKT and DKVMN despite substituting the binary readout with a GPCM head, confirming that the GPCM collapses to a standard logistic model when $K = 2$.

Table 2: Binary compatibility on ASSIST2015 and Synthetic-5 ($K = 2$). Best in bold.

Model	ASSIST2015		Synthetic-5	
	ACC	AUC	ACC	AUC
DKT	75.2%	0.727	**75.2%**	0.817
DKVMN	75.1%	**0.729**	75.1%	**0.827**
DEEP-GPCM	**75.4%**	0.723	74.7%	0.824

Learner State Dynamics (RQ2)

Figure 2 compares learner state trajectories across three models for four synthetic types of students. At each time t, we project θ_t to responses with the final output head of each model for the reference item. Therefore a faster convergence to the true projected category suggests higher confidence in modeling learning dynamics.

Specifically, dynamic GPCM's θ_t is a per-student embedding updated by a fixed recurrence, producing trajectories that are smooth but relatively flat, which means that the model adjusts slowly and converges to a stable estimate regardless of archetype. DEEP-GPCM and DKVMN+Softmax, sharing the same DKVMN backbone, produce trajectories by tracking the local response pattern, which yields fast convergence to projected responses even with the mid-ability type.

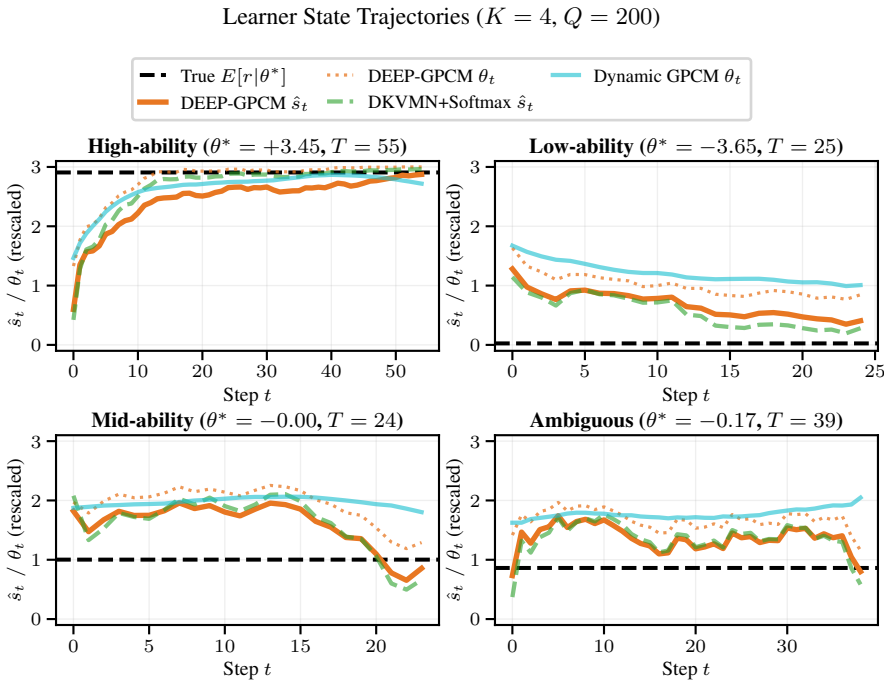

Figure 2: Learner state trajectories for four archetypes ($K = 4$, $Q = 200$). Curves smoothed with a 9-step rolling mean.

The similarity between DEEP-GPCM's θ_t and DKVMN+Softmax's expected score $\hat{s}_t$ is itself informative. Although both curves track each other closely across all four types, the trajectory reading from the GPCM head is able to output prediction on psychometrically interpretable scales, without losing the latent signal from the memory. The practical implication is that replacing a softmax head with a GPCM head costs nothing in trajectory quality while gaining insights to item-level characteristics, deriving richer context that $\hat{s}_t$ alone cannot support.

IRT Parameter Recovery and Stability (RQ3)

DEEP-GPCM recovers the discrimination parameter, α, more consistently than both GPCM baselines across most conditions ($r_\alpha = 0.75$–0.80 vs. 0.69–0.76 for Static and Dynamic GPCM; Table 3), despite receiving no direct supervision. This is suggests the DKVMN memory, trained solely on ordinal prediction, can distinguish item-level characteristics. Step difficulty recovery is slightly lower ($\bar{r}_\beta = 0.82$–0.92 for DEEP-GPCM vs. 0.97–0.99 for the baselines), which is expected as both GPCM bas dedicated lookup tables, offering an unobstructed gradient path to each β, whereas DEEP-GPCM infers thresholds as a byproduct of sequential prediction.

Figure 3 shows that the estimated ability distributions closely track the $\mathcal{N}(0, 1)$ generating prior, confirming population-level calibration. Low r_θ is expected, as the synthetic ground truth is a static scalar, whereas θ_t is a dynamic trajectory that updates with each interaction. A meaningful criterion for a dynamic estimator is whether θ_t, or projected response category converge at later interactions.

Figure 4 shows item parameter recovery. Static GPCM: $r_\alpha = 0.606$, $\bar{r}_\beta = 0.983$. Dynamic GPCM: $r_\alpha = 0.700$, $\bar{r}_\beta = 0.966$. DEEP-GPCM: $r_\alpha = 0.707$, $\bar{r}_\beta = 0.918$.

Table 3 reports the full breakdown across $K \in \{3, 4, 5\}$; taken together, DEEP-GPCM achieves competitive recovery of both item parameters without any direct IRT supervision.

Table 3: IRT parameter recovery across models and K ($Q = 200$, $N = 5000$). r_β is the mean across all $K-1$ thresholds.

K	Model	r_α	r_β (mean)	Per-threshold correlation
3	Static GPCM	0.732	**0.989**	r_{β_1}=0.990, r_{β_2}=0.987
	Dynamic GPCM	0.694	0.976	r_{β_1}=0.976, r_{β_2}=0.975
	DEEP-GPCM	**0.777**	0.816	r_{β_1}=0.814, r_{β_2}=0.818
4	Static GPCM	0.686	**0.983**	r_{β_1}=0.978, r_{β_2}=0.987, r_{β_3}=0.984
	Dynamic GPCM	**0.752**	0.966	r_{β_1}=0.958, r_{β_2}=0.973, r_{β_3}=0.967
	DEEP-GPCM	0.751	0.918	r_{β_1}=0.923, r_{β_2}=0.929, r_{β_3}=0.903
5	Static GPCM	0.759	**0.985**	r_{β_1}=0.980, r_{β_2}=0.987, r_{β_3}=0.991, r_{β_4}=0.982
	Dynamic GPCM	0.721	0.966	r_{β_1}=0.961, r_{β_2}=0.971, r_{β_3}=0.973, r_{β_4}=0.956
	DEEP-GPCM	**0.800**	0.912	r_{β_1}=0.913, r_{β_2}=0.929, r_{β_3}=0.922, r_{β_4}=0.885

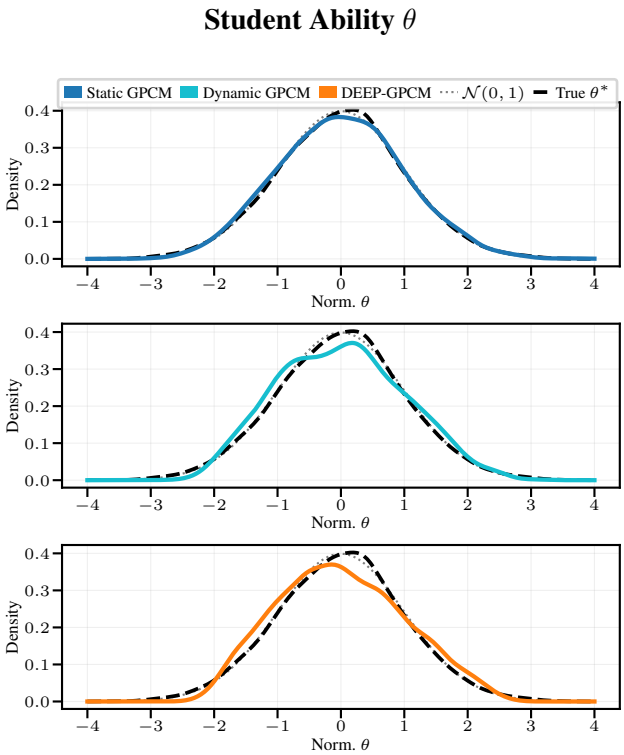

Figure 3: Student ability recovery ($K = 4$, $Q = 200$, $N = 5000$). Each row shows one model's estimated θ distribution compared to ground truth.

Scalable Item Encoding (RQ4)

We address **RQ4** by comparing three item encoding strategies on a large-scale dataset ($Q = 5000$, $K = 5$, sequences of length $[400, 600]$, 1000 students), where the cost of dense encoding becomes practically significant. Table 4 reports QWK and IRT parameter recovery for LinearDecay, Separable, and SIE encodings. All models use $d_k = d_v = 64$ and $N = 50$ memory slots; batch size 16 (reduced from 64 to accommodate LinearDecay's $\text{batch} \times S \times KQ$ input tensor). All three complete each epoch in 127–129 s, confirming that wall-clock throughput is governed by the DKVMN memory loop and not by the encoding stage. SIE achieves the best QWK (0.778) and the strongest discrimination recovery ($r_\alpha = 0.654$) at only 408K parameters — approximately one-fifth of LinearDecay's 1.9M — establishing SIE as the preferred encoding when both prediction quality and psychometric recovery matter at scale.

Ablation and Sensitivity Analysis

Loss components. We evaluate focal-only, WOL-only, and the combined loss (16) at $K = 4$. The combination ($\lambda_{\text{focal}} = \lambda_{\text{WOL}} = 0.5$) yields the strongest QWK. Removing focal loss primarily degrades accuracy on minority categories (highest rubric

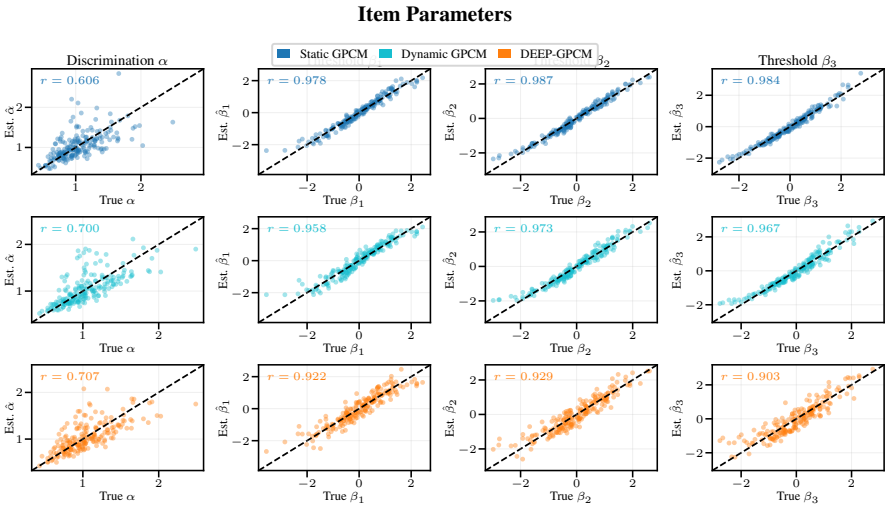

Figure 4: Item parameter recovery ($K = 4$, $Q = 200$, $N = 5000$). Each row shows one model; columns show discrimination α (log-linked) and step difficulties β_1–β_3 (z-scored). Pearson r is annotated per panel.

Table 4: Item encoding comparison at $Q = 5000$, $K = 5$. r_α, r_β: Pearson correlations with ground-truth IRT parameters (r_β pooled across thresholds).

Encoding	**Params**	**QWK**	r_θ	r_α	r_β
LinearDecay	1943K	0.762	0.14	0.022	0.906
Separable	667K	0.773	0.22	0.246	0.911
SIE (ours)	408K	**0.778**	0.23	**0.654**	0.884

levels), confirming its value for class-imbalanced partial-credit distributions. Removing the ordinal distance penalty γ from WOL reduces Kendall's τ without much effect on accuracy, isolating its contribution to rank-order consistency.

Table 5: Ablation of loss components at $K = 4$ (single seed, $Q = 200$, $N = 5000$).

Loss variant	**QWK**	τ
Full (λ_{focal}=λ_{WOL}=0.5)	0.683	0.513
Focal only (λ_{focal}=1, λ_{WOL}=0)	0.679	0.508
WOL only (λ_{focal}=0, λ_{WOL}=1)	0.684	0.516

Monotonic threshold constraint. To isolate the contribution of the softplus gap construction (13), we conduct a controlled ablation at $K = 5$ with three conditions: (1) the baseline DEEP-GPCM model with monotonic thresholds and ordinal losses (focal +

WOL), (2) an unconstrained variant that replaces the softplus gap construction with a single linear layer $\boldsymbol{\beta} = W_\beta \mathbf{q} + \mathbf{b}_\beta$ producing $K-1$ thresholds without ordering guarantees, trained with ordinal losses, and (3) the unconstrained variant trained with pure cross-entropy loss (focal_weight $= 0$, weighted_ordinal_weight $= 0$) to test whether ordinal losses or the GPCM likelihood itself enforce ordering. All three models use identical architectures (Static Item Embedding, $N = 5000$, $Q = 200$, 15 epochs, seed 42) and differ only in the threshold parameterisation and loss composition.

Results in Table 6 show that removing the monotonicity constraint produces near-zero disordered step difficulties when trained with ordinal losses (0.0% disorder), but 0.5% disorder emerges without ordinal losses. Inspection of the learned parameters reveals that the unconstrained linear layer's bias vector converges to approximately ordered values even without ordinal losses (e.g., $[-0.90, -0.25, 0.25, 0.89]$), indicating that the GPCM likelihood provides weak implicit ordering pressure. However, ordinal losses are necessary to fully eliminate disorder. QWK degrades marginally when removing the monotonicity constraint ($\Delta\text{QWK} \approx -0.002$ with ordinal losses, -0.02 without), but parameter recovery degrades substantially: $\bar{r}_\beta$ drops from 0.912 (monotonic) to 0.694 (unconstrained + ordinal) to 0.545 (unconstrained + no ordinal), and r_α similarly degrades from 0.760 to 0.600 to 0.656. This indicates that while unconstrained thresholds can achieve comparable predictive performance, they sacrifice interpretability — the learned $\boldsymbol{\beta}$ vectors no longer faithfully represent the ground-truth item step difficulties, undermining the psychometric validity of the extracted parameters.

Table 6: Monotonicity constraint ablation at $K = 5$ ($Q = 200$, $N = 5000$, 15 epochs, seed 42). Unconstrained models replace the softplus gap construction with a single linear layer producing $K-1$ thresholds without ordering guarantees. Recovery correlations computed on the test set.

Model	**ACC**	**QWK**	$\boldsymbol{\tau}$	$\boldsymbol{r_\alpha}$	$\boldsymbol{\bar{r}_\beta}$	**Disorder (%)**
Monotonic + ordinal	0.511	0.759	0.576	0.760	0.912	0.0
Unconstrained + ordinal	0.507	0.757	0.574	0.600	0.694	0.0
Unconstrained + NO ordinal	0.513	0.736	0.549	0.656	0.545	0.5

Sensitivity analyses. We evaluate robustness to category resolution $K \in \{3, 4, 5\}$ (results reported in Table 1), class imbalance severity, and interaction sequence length. Ordinal agreement (QWK) degrades gracefully under mild to moderate class skew, with the focal loss component absorbing much of the imbalance. Shorter sequences ($S < 50$) reduce ability recovery quality, as the model has insufficient interaction history to converge θ_t; this is consistent with minimum-length requirements in classical IRT estimation. Robustness to severe imbalance and very short sequences ($S < 20$) remains an open question pending additional experimentation.

Table 5 reports QWK and τ for the three loss variants at $K = 4$ (single seed). Differences are within $\Delta\text{QWK} < 0.006$ and $\Delta\tau < 0.008$, indicating that the combined loss provides a consistent but modest improvement on this synthetic dataset.

We evaluate sequence length sensitivity at $K = 4$ across three regimes: $S \in [20, 80]$ (baseline), $S \in [20, 100]$, and $S \in [20, 40]$. QWK follows the expected direction (0.6903, 0.6825, 0.6742 respectively), with a modest drop under very short sequences. Item-level recovery ($\bar{r}_\beta \approx 0.92$ across all regimes) is stable, consistent with the person–item identification asymmetry: item parameters are estimated from $N = 5000$ student interactions and are robust to individual sequence length, whereas discrimination recovery (r_α) degrades more markedly from 0.786 to 0.524 as sequences shorten and item co-occurrence patterns become sparser. Ability recovery (r_θ) is similarly lower at $S \in [20, 40]$ (0.158 vs. 0.191 baseline), reflecting the reduced interaction history available per student. Table 7 reports the full results.

Table 7: Sequence length sensitivity at $K = 4$ ($Q = 200$, $N = 5000$, seed 42, 15 epochs). Recovery correlations computed on the test set.

Regime	**ACC**	**QWK**	τ	r_θ	r_α	$\bar{r}_\beta$
$S \in [20, 80]$ (baseline)	0.529	0.683	0.513	0.191	0.715	0.918
$S \in [20, 100]$	0.530	0.690	0.518	0.168	0.786	0.927
$S \in [20, 40]$	0.523	0.674	0.505	0.158	0.524	0.916

Proxy-Ordinality on Binary Benchmarks (RQ5)

Addressing **RQ5**, we evaluate whether ordinal modeling gains persist when response categories must be constructed from proxy signals rather than purpose-designed rubrics. Real KT benchmarks are predominantly binary, yet process features such as attempt counts, hint requests, and response latency encode graded evidence of understanding (Lu et al., 2024). We derive $K = 3$ ordered categories from attempt counts in ASSISTments-style interaction logs: 1 attempt (= mastery), 2–3 attempts (= partial understanding), 4+ attempts (= struggle). This mapping approximates an ordinal rubric but introduces noise — guessing, rapid retries, and help-seeking behavior may violate the assumed ordering. We therefore treat this setting as a *stress test* of the model's ordinal structure under imperfect category construction, and ask whether the GPCM head exploits the constructed step transitions (supporting RQ5 positively) or whether category noise erases the gains (informative null result).

Results for DEEP-GPCM variants against DKVMN (softmax) are reported in Table 8, using QWK as the primary metric. The table is complete once the ASSISTments pipeline is run; the expected pattern and sensitivity analysis are described in the note below.

Pending: proxy-ordinality experiments

Running the ASSISTments attempt-count converted proxy-oridnal case: DEEP-GPCM should show QWK gains over DKVMN (softmax) if ordinal category construction is consistent enough for the GPCM formulation to exploit step transitions; if attempt-count bins are too noisy, the QWK gap may narrow.

Table 8: Proxy-ordinality results on real educational logs. Ordered categories from attempt count bins ($K = 3$). QWK is the primary metric.

Model	**Acc(%)**	**QWK**	τ
DKVMN (softmax)	—	—	—
DEEP-GPCM (LinearDecay)	—	—	—
DEEP-GPCM (SIE)	—	—	—

Discussion

Educational Implications

DEEP-MIRT extends binary KT to address wider applications for formative assessment practice. By estimating a dynamic ability and categorical probabilities, it differentiates partial knowledge from total misunderstanding, providing richer insights in continuous learning progressions. Furthermore, the extracted item parameters describe item-level characteristics beyond immediate predictive use. Unlike classical psychometric calibration, DEEP-MIRT allows online extraction of item characteristics alongside student states, which helps identify underperforming or misaligned items in real time. This transition from a purely predictive instrument to a measurement-aware system ensures that instructional decisions are grounded in identifiable, theoretically-backed constructs such as discrimination and step difficulty (Heritage, 2007; Wilson, 2005).

Limitations and Future Directions

While the framework establishes a foundation for polytomous knowledge tracing, several limitations present direct avenues for future research.

First, although we proposed DEEP-MIRT as a modular framework with interchangeable components, the current implementation DEEP-GPCM is limited to the GPCM head. A significant research opportunity lies in fully realizing this modularity. For example, a nominal response head using the NRM (Bock, 1972) would extend the framework to unordered polytomous responses such as multiple-choice distractors, enabling misconception-level feedback.

Second, evaluating parameter recovery on synthetic data generated under GPCM assumptions is inherently circular, and proxy signals like attempt counts contain construct-irrelevant variance. Future evaluations will apply the model to authentic polytomous datasets and validate the inferred trajectories against external standardized measures.

Third, while the current backbone effectively maintains a unidimensional student state but struggles with multi-dimensional discrimination recovery. Connecting multi-dimensional instantiations with concept-aligned memory slots would enable per-skill ability tracking, supporting more targeted tracking than a composite θ_t.

Conclusion

The transition from binary to polytomous responses is essential for practical educational assessment. DEEP-MIRT reconciles the predictive performance of sequential neural networks with the rigorness of IRT. By decoupling the sequential backbone from interchangeable psychometric heads, we demonstrate that deep learning can optimize prediction while yielding identifiable measurement constructs. The framework's backward compatibility and scalability confirm its viability as a practical replacement for existing systems. By respecting the ordinal structure of educational data, DEEP-MIRT establishes an alternative for measurement-aware adaptive learning.

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